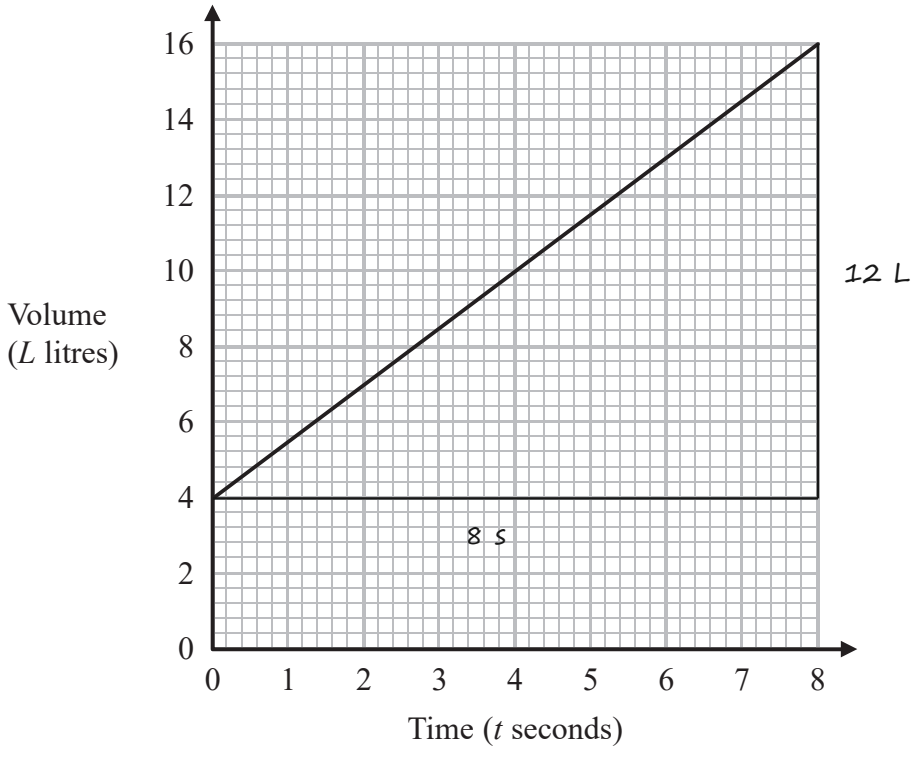
12 The graph shows the volume of liquid (L litres) in a container at time t seconds.



(a) Find the gradient of the graph.

Gradient = $\frac{16 - 4}{8 - 0} = \frac{12}{8} = 1.5$

1 mark
Use the changes on both axes

1 mark
Correct gradient

1.5

(2)

(b) Explain what this gradient represents.

The volume increases by 1.5 litres every second.

1 mark
Explain the rate of filling

(1)

The graph intersects the volume axis at $L = 4$

(c) Explain what this intercept represents.

There are 4 litres in the container at the start, when t = 0.

1 mark
Interpret the initial volume

(1)

(Total for Question 12 is 4 marks)